government orders all its citizens to have their DNA scanned and to share all their medical data with some central authority, it would gain an immense advantage in genetics and medical research over societies in which medical data is strictly private. The main handicap of authoritarian regimes in the twentieth century – the attempt to concentrate all information in one place – might become their decisive advantage in the twenty-first century.

As algorithms come to know us so well, authoritarian governments could gain absolute control over their citizens, even more so than in Nazi Germany, and resistance to such regimes might be utterly impossible. Not only will the regime know exactly how you feel – it could make you feel whatever it wants. The dictator might not be able to provide citizens with healthcare or equality, but he could make them love him and hate his opponents. Democracy in its present form cannot survive the merger of biotech and infotech. Either democracy will successfully reinvent itself in a radically new form, or humans will come to live in 'digital dictatorships'.

This will not be a return to the days of Hitler and Stalin. Digital dictatorships will be as different from Nazi Germany as Nazi Germany was different from *ancien régime* France. Louis XIV was a centralising autocrat, but he did not have the technology to build a modern totalitarian state. He suffered no opposition to his rule, yet in the absence of radios, telephones and trains, he had little control over the day-to-day lives of peasants in remote Breton villages, or even of townspeople in the heart of Paris. He had neither the will nor the ability to establish a mass party, a countrywide youth movement, or a national education system.[30] It was the new technologies of the twentieth century that gave Hitler both the motivation and the power to do such things. We cannot predict what will be the motivations and powers of digital dictatorships in 2084, but it is very unlikely that they will just copy Hitler and Stalin. Those gearing themselves up to refight the battles of the 1930s might be caught off their guard by an attack from a totally different direction.

Even if democracy manages to adapt and survive, people might become the victims of new kinds of oppression and discrimination. Already today more and more banks, corporations and institutions are using algorithms to analyse data and make decisions about us. When you apply to your bank for a loan, it is likely that your application is processed by an algorithm rather than by a human. The algorithm analyses lots of data about you and statistics about millions of other people, and decides whether you are reliable enough to give you a loan. Often, the algorithm does a better job than a human banker. But the problem is that if the algorithm discriminates against some people unjustly, it is difficult to know that. If the bank refuses to give you a loan, and you ask 'Why?', the bank replies 'The algorithm said no.' You ask 'Why did the algorithm say no? What's wrong with me?', and the bank replies 'We